

Title

Resonance Spectrum Mathematics for Collapse Proximity: An Operator-Geometric Specification of $D(t)$, $\widehat{\text{PSCI}}$, and Spectral Early Warning

Subtitle

A preregisterable mathematical framework linking resonance, coherence loss, non-normal amplification, and critical spectral contraction across open physiological systems

Keywords

Spectral collapse, resonance, operator geometry, non-self-adjoint dynamics, pseudospectrum, biorthogonal spectrum, Petermann factor, critical scaling, coherence, disruption index, $D(t)$, $\widehat{\text{PSCI}}$, Lindblad dynamics, early warning, non-normal amplification, multichannel physiology, spectral entropy, critical manifold, collapse susceptibility

Abstract

This document defines a compact mathematical framework for resonance-centered spectral collapse analysis in open, multichannel physiological systems. The core construction treats system evolution as governed by a generally non-self-adjoint operator L acting on a separable Hilbert space H , with instability encoded in the spectrum $\sigma(L)$, the resolvent $R(z;L)=(L-zI)^{-1}$, and a stability functional $V(L)=\inf_{\lambda \in \sigma(L)} \text{Re} \lambda$. Collapse is defined structurally by approach to the critical manifold $M_c=\{L: V(L)=0\}$.

The framework introduces a resonance interpretation of instability, in which forced response, non-normal amplification, pseudospectral inflation, and coherence loss are treated as measurable shadows of spectral reorganization. A practical bridge is then defined from operator quantities to computable surrogates from time series: cross-channel coherence $C(t)$, disruption index $D(t)=[C_0-C(t)]/C_0$, local instability proxy $\lambda^{\max}(t)$, spectral entropy aggregate $H^{\text{spec}}(t)$, fractal coalescence $\text{Var}_\alpha(t)$, and a monotone proximity index $\widehat{\text{PSCI}}(t)$.

The central hypothesis is that resonance near collapse is governed not only by eigenvalue proximity to the imaginary axis, but also by non-normal amplification, quantified through biorthogonal geometry and pseudospectral growth. Under this interpretation, rising $D(t)$

and $\widehat{\mathrm{PSCI}}(t)$ are treated as operational indicators of approach to collapse. Critical versus non-critical behavior is discriminated through preregistered likelihood-based model comparison between linear decay and critical scaling

$$D(t) = k(tc - t) + \epsilon D(t) = k(t_c - t)^\nu + \epsilon D(t) = k(tc - t) + \epsilon t.$$

The contribution is not a claim of validated invariance, but a definition-aligned mathematical specification: a way to express resonance, coherence loss, and collapse proximity within a single operator-geometric language that is computable, falsifiable, and extensible to open-system physics, biology, and multiscale dynamical inference.

Introduction

Resonance is usually treated as a response phenomenon: a system is driven, a frequency is matched, amplitude rises. Collapse is usually treated as a failure phenomenon: a system destabilizes, coherence degrades, structure gives way. This document proposes that these are not separate categories, but adjacent views of the same spectral process.

The central idea is simple:

- Resonance is the observable expression of spectral proximity.
- Collapse is the structural consequence of spectral instability.
- Coherence loss is one measurable projection of that transition.
- Early warning becomes possible when resonance sharpens before collapse completes.

To formalize that idea, this text defines a spectral mathematics in which an open system is represented by a generally non-self-adjoint generator L on a Hilbert space H . The system's behavior is then read through:

- its spectrum $\sigma(L)$,
- its resolvent $R(z; L)$,
- its least-stable direction $V(L)$,
- its pseudospectral sensitivity,
- and its observable surrogates in time-series data.

The emphasis herein is on resonance, because resonance is where instability first becomes audible to measurement. A mode does not need to cross into overt divergence to matter; it may already dominate response through narrowing damping, non-orthogonal amplification, or pseudospectral swelling. In that sense, resonance is the measurable edge of collapse.

This specification is meant to serve four purposes:

1. Mathematical: define the spectral objects clearly and minimally.
2. Operational: identify computable surrogates that can be preregistered.
3. Epistemic: separate structural claims from empirical validation claims.
4. Integrative: provide a common language for physics, biology, and measurement science.

The result is not a finished theorem of nature, but a disciplined scaffold: a way to speak about resonance-associated danger knowledge, especially through $D(t)D(t)D(t)$, without collapsing prematurely into metaphor, risk scoring, or unverifiable universals.

Body of Evidence

1. Operator, spectrum, and spectral measure

Let H be a separable Hilbert space and let

$$L: D(L) \subset H \rightarrow H : D(L) \subset H \rightarrow H$$

be a closed, possibly non-self-adjoint generator.

Define:

- Spectrum: $\sigma(L) \subset \mathbb{C} \setminus \sigma(L) \subset \mathbb{C}$
- Resolvent set: $\rho(L) = \mathbb{C} \setminus \sigma(L)$
- Resolvent: $R(z; L) = (L - zI)^{-1}, z \in \rho(L)$

In the generalized biorthogonal case,

$$L = \int \lambda d\Pi(\lambda) L = \int \lambda d\Pi(\lambda)$$

with non-orthogonal spectral family satisfying, in general,

$$\Pi(\lambda)\Pi(\mu) \neq \delta(\lambda - \mu)\Pi(\lambda)\Pi(\mu) \neq \delta(\lambda - \mu)\Pi(\lambda)$$

For a state $\psi \in H$, define the induced spectral measure

$$\mu_\psi(B) = \langle \psi, E(B)\psi \rangle$$

for Borel subsets $B \subset \sigma(L)$.

2. Spectral distance to instability

Define the stability functional

$$V(L) = \inf_{\lambda \in \sigma(L)} \operatorname{Re} \lambda$$

This gives the least-stable real spectral direction.

Define the critical manifold

$$M_c = \{L: V(L) = 0\} \quad M_c = \{L: V(L) = 0\} \quad M_c = \{L: V(L) = 0\}$$

This is the structural boundary between damped and marginally unstable regimes.

Define the spectral gap

$$\Delta\lambda(L) = \min\{|\lambda_i - \lambda_j| : \lambda_i \neq \lambda_j \in \sigma(L)\} \quad \Delta\lambda(L) = \min\{|\lambda_i - \lambda_j| : \lambda_i \neq \lambda_j \in \sigma(L)\}$$

when point spectrum is defined.

Define the pseudospectral sensitivity set

$$\Lambda_\varepsilon(L) = \{w \in \mathbb{C} : \|(L - wI)^{-1}\| > \varepsilon^{-1}\} \quad \Lambda_\varepsilon(L) = \{w \in \mathbb{C} : \|(L - wI)^{-1}\| > \varepsilon^{-1}\}$$

and the pseudospectral proximity

$$d_\varepsilon(L) = \inf\{|z - w| : z \in \sigma(L), w \in \Lambda_\varepsilon(L)\} \quad d_\varepsilon(L) = \inf\{|z - w| : z \in \sigma(L), w \in \Lambda_\varepsilon(L)\}$$

This captures non-normal sensitivity not visible from eigenvalues alone.

3. Resonance under forcing

Consider a forced linear system

$$d\psi/dt = L\psi + fe^{i\omega t} \quad d\psi/dt = L\psi + fe^{i\omega t}$$

Then the steady-state response amplitude is

$$\psi^*(\omega) = (i\omega I - L)^{-1} b \cdot f \quad \psi^*(\omega) = (i\omega I - L)^{-1} b \cdot f$$

For observable vector c , define the transfer function

$$H(\omega) = \langle c, (i\omega I - L)^{-1} b \rangle \quad H(\omega) = \langle c, (i\omega I - L)^{-1} b \rangle$$

In biorthogonal modal form,

$$H(\omega) = \sum_k \langle c, \psi_k \rangle \langle \psi_k, b \rangle / (i\omega - \lambda_k) \quad H(\omega) = \sum_k \langle c, \psi_k \rangle \langle \psi_k, b \rangle / (i\omega - \lambda_k)$$

This shows directly that resonant amplification occurs near poles of the resolvent, that is, when $i\omega$ approaches the spectrum.

Operationally:

- resonance sharpens as $\text{Re}\lambda_k \rightarrow 0^-$ - $\text{Re}\lambda_k \rightarrow 0^-$,
- amplitude increases as damping narrows,
- and non-normality can magnify the response even before instability is overt.

4. Non-normal amplification

Let right and left eigenvectors satisfy

$$L\psi_k R = \lambda_k \psi_k R \quad L\psi_k^* R = \bar{\lambda}_k \psi_k^* R \quad L\psi_k R = \lambda_k \psi_k R$$

and

$$L^\dagger \psi_k L = \bar{\lambda}_k \psi_k L \quad L^\dagger \psi_k^* L = \bar{\lambda}_k \psi_k^* L \quad L^\dagger \psi_k L = \bar{\lambda}_k \psi_k L$$

with normalization

$$\langle \psi_j L, \psi_k R \rangle = \delta_{jk} \quad \langle \psi_j^* L, \psi_k^* R \rangle = \delta_{jk} \quad \langle \psi_j L, \psi_k R \rangle = \delta_{jk}$$

Define the Petermann factor

$$K_k = \frac{\langle \psi_k L, \psi_k R \rangle \langle \psi_k^* L, \psi_k^* R \rangle}{|\langle \psi_k L, \psi_k^* R \rangle|^2} \geq 1 \quad K_k = \frac{|\langle \psi_k L, \psi_k^* R \rangle|}{|\langle \psi_k L, \psi_k R \rangle| |\langle \psi_k^* L, \psi_k^* R \rangle|} \geq 1$$

This measures eigenvector non-orthogonality.

Define the approximate single-mode susceptibility

$$\chi_k(\omega) \approx K_k |i\omega - \lambda_k|^{-1} \quad \chi_k(\omega) \approx \frac{K_k}{i\omega - \lambda_k}$$

Thus resonance is amplified not only by spectral proximity but by biorthogonal ill-conditioning. This is essential when resonance is interpreted as a precursor rather than merely a peak.

5. Spectral collapse and entropy

Let $\rho(\lambda; t)$ denote a spectral density.

Define spectral contraction toward a critical mode λ_c by weak-* convergence

$$\rho(\lambda; t) \rightarrow \delta(\lambda - \lambda_c) \quad \text{as } t \rightarrow t_c^- \quad \rho(\lambda; t) \rightarrow \delta(\lambda - \lambda_c) \quad \text{as } t \rightarrow t_c^-$$

Define the spectral entropy

$$S_{\text{spec}}(t) = -\int \sigma(L) \rho(\lambda; t) \log \rho(\lambda; t) d\lambda \quad S_{\text{spec}}(t) = -\int \sigma(L) \rho(\lambda; t) \log \rho(\lambda; t) d\lambda$$

Under open-system contraction, the expected trend is

$$dS_{\text{spec}}/dt \leq 0 \quad \text{for } dS_{\text{spec}}/dt \leq 0$$

Define the regulated collapse functional

$$C_\epsilon[L] = \int \sigma(L) |\lambda - \lambda_c| - \epsilon \rho(\lambda) d\lambda, 0 < \epsilon < 1 \quad C_\epsilon[L] = \int \sigma(L) |\lambda - \lambda_c| - \epsilon \rho(\lambda) d\lambda, 0 < \epsilon < 1$$

Collapse is signaled by

$$C_\epsilon[L] \rightarrow \infty \quad C_\epsilon[L] \rightarrow \infty$$

6. Critical scaling

Let a control parameter $g \mapsto L(g)$ be defined, with least-stable branch $\lambda_{\min}(g)$ satisfying

$$\text{Re} \lambda_{\min}(g) \downarrow 0 \text{ as } g \uparrow g_c \quad \text{Re} \lambda_{\min}(g) \downarrow 0 \text{ as } g \uparrow g_c$$

Assume gap scaling

$$\Delta \lambda(g) \sim A(g_c - g)^\nu, \nu > 0 \quad \Delta \lambda(g) \sim A(g_c - g)^\nu, \nu > 0$$

Then idealized resonance sharpness behaves as:

- peak height: $\sim 1/\text{Re}(-\lambda_{\min}(g)) \sim (g_c - g)^{-\nu} \sim \frac{1}{(g_c - g)^\nu} \sim \frac{1}{(g_c - g)^\nu}$
- peak width: $\sim \text{Re}(-\lambda_{\min}(g)) \sim (g_c - g)^\nu \sim \frac{1}{(g_c - g)^\nu} \sim \frac{1}{(g_c - g)^\nu}$

Thus critical resonance becomes sharper and taller as the system approaches structural instability.

7. Data-to-operator surrogates

Because LLL is rarely observed directly in physiological applications, define computable surrogates.

Cross-channel coherence

For K channels and pairwise magnitude-squared coherence $\gamma_{ij}^2(f)$ over bands $B = \{b_m\}$,

$$\bar{C} = \sum_m a_m \cdot \text{median}_{i < j} \{ \text{mean}_{f \in b_{mij}}(f) \} \quad \bar{C} = \sum_m a_m \cdot \text{median}_{i < j} \{ \text{mean}_{f \in b_{mij}}(f) \}$$

with $a_m \geq 0$ and $\sum_m a_m = 1$.

Disruption index

$$D(t) = \frac{C_0 - C(t)}{C_0}$$

where C_0 is baseline coherence.

This is the key operational form of knowledge at danger: a normalized measure of how much system-level coordination has degraded from reference.

Local instability proxy

$$\lambda^{\max}(t)$$

estimated from short-time Lyapunov methods and aggregated across channels.

Spectral entropy aggregate

$$\bar{H}^{\text{spec}}(t) = \text{median}_k \{ H_k(t) \} \quad \bar{H}^{\text{spec}}(t) = \text{median}_k \{ H_k(t) \}$$

Fractal coalescence

$$\text{Var}_\alpha(t) = \text{Var}_k(t) \quad \text{Var}_\alpha(t) = \text{Var}_k(t)$$

8. Monotone proximity index

Define the Pellis Spectral Collapse Index surrogate

$$\text{PSCI}(t) = w_1 z(1 - \bar{C}(t)) + w_2 z(-\bar{H}^{\text{spec}}(t)) + w_3 z(\lambda^{\max}(t)) + w_4 z(-\text{Var}_\alpha(t))$$

where:

- $z(\cdot)$ is baseline z-scoring,
- $w_j \geq 0$,
- $\sum_j w_j = 1$.

Monotonicity claim:

If

$$V(L(t)) = \inf \operatorname{Re} \lambda V(L(t)) = \inf \operatorname{Re} \lambda$$

decreases toward zero and non-normal amplification rises, then

$$\widehat{\operatorname{PSCI}}(t) \operatorname{PSCI}(t)$$

should increase.

This gives a practical scalar for collapse proximity.

9. Resonance–coherence linkage

For a linear stochastic system

$$d\psi = L\psi dt + \Sigma dW_t \quad \psi = L\psi dt + \Sigma dW_t$$

with outputs

$$y = C\psi \quad y = C\psi$$

the cross-spectral density is

$$\begin{aligned} S_{yy}(\omega) &= C(i\omega I - L)^{-1} \Sigma \Sigma^\dagger (-i\omega I - L^\dagger)^{-1} C^\dagger S_{yy}(\omega) = C(i\omega I - L)^{-1} \\ &\Sigma \Sigma^\dagger (-i\omega I - L^\dagger)^{-1} C^\dagger \\ C^\dagger S_{yy}(\omega) &= C(i\omega I - L)^{-1} \Sigma \Sigma^\dagger (-i\omega I - L^\dagger)^{-1} C^\dagger \end{aligned}$$

and pairwise coherence is

$$\gamma_{ij}^2(\omega) = \frac{|S_{ij}(\omega)|^2 S_{ii}(\omega) S_{jj}(\omega)}{|S_{ii}(\omega)|^2 |S_{jj}(\omega)|^2} \quad \gamma_{ij}^2(\omega) = \frac{|S_{ij}(\omega)|^2}{|S_{ii}(\omega)|^2 |S_{jj}(\omega)|^2}$$

As $\operatorname{Re} \lambda_{\min} \rightarrow 0^+$, and different outputs project unequally onto the inflating least-stable mode, band-averaged coherence may decline, giving

$$D(t) \uparrow D(t) \uparrow$$

This provides the toy-model rationale for treating $D(t)D(t)D(t)$ as a gap-closure proxy.

10. Decay-form discrimination

Critical model:

$$D(t) = k(t_c - t)^\nu + \epsilon t \quad D(t) = k(t_c - t)^\nu + \epsilon t$$

with

- $k > 0$
- $\nu > 0$
- $t_c \in (t_{\text{end}}, t_{\text{end}} + \Delta]$

Linear model:

$$D(t) = \alpha + \beta t + \epsilon$$

Decision rule:

- choose critical if $\Delta \text{WAIC} = \text{WAIC}_{\text{lin}} - \text{WAIC}_{\text{crit}} \geq 4$ and the 95% CI for ν lies in $(0, \infty)$
- choose linear if $\Delta \text{WAIC} \leq -4$ and the 95% CI for ν lies in $(0, \infty)$
- otherwise inconclusive.

11. Open-system contraction

Density operator evolution is given by GKSL form:

$$\frac{d\rho}{dt} = D[\rho] = -i[H, \rho] + \sum_k (L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\})$$

This induces spectral contraction and supports the expectation

$$\frac{dS_{\text{spec}}}{dt} \leq 0$$

with approach to collapse corresponding to

$$V(L(t)) \rightarrow 0 + V(L(t)) \rightarrow 0 +$$

and concentration of spectral mass.

12. Pseudospectral early warning

Define pseudospectral radius at the imaginary axis

$$\text{pps}(i\omega; \epsilon) = \sup \{ \| (i\omega I - L) - 1 \| : i\omega \in \Lambda_\epsilon(L) \}$$

Define early warning metric

$$\Xi(t) = \max_{\omega \in B} \| (i\omega I - L(t)) - 1 \|$$

where $L^{\wedge}(t)\hat{L}(t)L^{\wedge}(t)$ is an identified local linear model.

As pseudospectral contours expand toward the imaginary axis, $\Xi(t)\Xi(t)\Xi(t)$ rises. This complements $PSCI^{\wedge}(t)\widehat{\mathrm{PSCI}}(t)PSCI(t)$ by tracking resonance danger directly in operator-response space.

Conclusions

The novelty of this framework is not that it mentions resonance, coherence, or instability separately. Those are known. The novelty is that it binds them into one mathematical object-chain:

$$L \rightarrow \sigma(L) \rightarrow R(z; L) \rightarrow V(L), \Delta\lambda, Kk, \Xi \rightarrow C(t), D(t), PSCI^{\wedge}(t) L \rightarrow \sigma(L) \rightarrow R(z; L) \rightarrow V(L), \Delta\lambda, Kk, \Xi \rightarrow C(t), D(t), PSCI(t)$$

That chain matters because it changes the use of measurement.

Instead of asking only:

- "Is the system at risk?"
- "Did a classifier detect a pattern?"
- "Did one channel change?"

the framework asks:

- "Is the system becoming more resonant in an instability-relevant way?"
- "Is coherence loss a projection of operator reorganization?"
- "Is the system approaching a mathematically defined collapse boundary?"

In that sense, Knowledge = $D(t)D(t)D(t)$ @Danger means:

- $D(t)D(t)D(t)$ is not merely loss,
- it is structured foreknowledge of instability,
- especially when interpreted jointly with resonance sharpening, non-normal amplification, and pseudospectral inflation.

Applied benefit follows directly:

- earlier detection of approach-to-collapse,
- better distinction between noise and real precursor structure,
- and a path from physiology to physics without pretending the bridge is already proven.

This is a measurement framework with mechanistic ambition and falsifiable restraint.

Historical Notes

This specification emerges from an attempt to unify several strands that have often remained separate:

- classical spectral operator theory,
- non-Hermitian and pseudospectral dynamics,
- resonance response theory,
- open-system entropy contraction,
- multichannel physiological coherence analysis,
- and critical-scaling inference.

Its distinctive orientation is post-feature, pre-dogma:

- after naive feature engineering,
- before universality is declared.

It is historically situated in a phase of science where early-warning claims are common, but rigorous operator-grounded bridges between biology and instability geometry remain rare. Its emphasis on surrogate firewalls, non-normality, and preregisterable surrogates reflects that context.

Credits

Primary conceptual direction: ERES–Pellis resonance/collapse alignment

Spectral operator framing: operator geometry, non-self-adjoint dynamics, resolvent analysis

Measurement bridge: multichannel coherence, disruption index $D(t)D(t)D(t)$, and collapse-proximity surrogate $\widehat{\mathrm{PSCI}}(t)$

Protocol logic: identifiability first, criticality second, correspondence third

Intended interlocution: mathematical physics, open-system theory, physiological dynamics, and resonance-grounded collapse inference

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